

Lesson 12 — Loops with Motion & Sound (+ Debug)

Matches LEGO Teacher Portal lesson: "Dragon Defense".

Objective (I can...): I can put several actions inside one loop so a pattern repeats. **CS concept:** Multi-action loop body · **Builds on:** Lesson 11 (single-action repeat).

AI Connection (2 min): Each training "round" repeats the same steps on new examples until the AI improves. *First-principles question:* "What might get better each time the loop runs again?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (8): Pattern clap — coach does "clap-stomp-clap-stomp"; pairs find the repeating chunk that could go in a loop. - **Concept + demo (10):** Build a loop body with move + sound + wait; run it and point out the whole chunk repeats. - **Build/Code (30):** Pairs code a 3-action repeating pattern inside one loop (spin → beep → wait). **Debug mini-task (3 min):** hand each pair a "broken" program where a block sits *outside* the loop — they find and fix it so the whole pattern repeats. *Swap at 15 min.* - **Challenge (7):** Add a light or reverse move to the loop body; re-run. - **Share + pack-up (5):** Pairs demo the pattern and name the actions inside the loop. Model stays built.

Pair roles: Driver codes the loop body, Navigator lists the actions to include. Swap at 15 min. (*Trio: third leads the debug hunt.*)

Differentiation: *Support* — a 2-action loop body with a twin example. *Extension / fast-finisher* — a 4-action loop body. *Reteach* — build the body outside a loop first, confirm it works, then drag it inside.

Exit check: Pair runs a loop whose body has 3+ actions that repeat together. **At-home (optional):** Make up a 3-move dance; repeat it 4 times. ~10 min.
